Networks

Multiplication

Derived networks

Fractional approach

Normalizations

Example

Conclusions

References

Variations on projection of 2-mode network

Vladimir Batagelj

IMFM Ljubljana and IAM UP Koper

COMPSTAT 2026

The 27th International Conference on Computational Statistics

Athens University Law School, 25-28 August 2026

Athens, Greece

Networks

Multiplication

Derived networks

Fractional approach

Normalizations

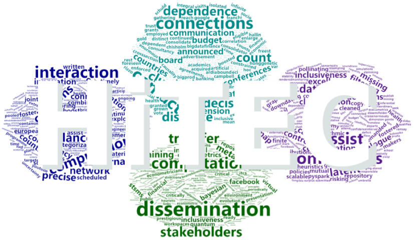
Example

Conclusions

References

Outline

1 Networks
2 Multiplication
3 Derived networks
4 Fractional approach
5 Normalizations
6 Example
7 Conclusions
8 References



Vladimir Batagelj: `vladimir.batagelj@fmf.uni-lj.si`

Current version of slides (August 27, 2026 at 23 : 09): slides PDF

`https://github.com/bavla/NormNet/tree/main/TwoMode`

Networks

Network = Graph + Data

A *network* $\mathcal{N} = (\mathcal{V}, \mathcal{L}, \mathcal{P}, \mathcal{W})$ consists of:

- a *graph* $\mathcal{G} = (\mathcal{V}, \mathcal{L})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{A}$ is the set of arcs (directed links), $\mathcal{E}$ is the set of edges (undirected links), and $\mathcal{L} = \mathcal{E} \cup \mathcal{A}$ is the set of links. $n = |\mathcal{V}|$, $m = |\mathcal{L}|$
- $\mathcal{P}$ *node value functions* / properties: $p : \mathcal{V} \to A$
- $\mathcal{W}$ *link value functions* / weights: $w : \mathcal{L} \to B$

In a *two-mode* network $\mathcal{N} = ((\mathcal{U}, \mathcal{V}), \mathcal{L}, \mathcal{P}, \mathcal{W})$ the set of nodes consists of two disjoint sets of nodes $\mathcal{U}$ and $\mathcal{V}$, and all the links from $\mathcal{L}$ have one end node in $\mathcal{U}$ and the other in $\mathcal{V}$.

$$\mathrm{wod}(v) = \sum_{e \in \mathrm{ostar}(v)} w(e) \quad \textit{and} \quad \mathrm{wid}(v) = \sum_{e \in \mathrm{istar}(v)} w(e)$$

where $\mathrm{ostar}(v)$ is the out-star – links leading out of node v, etc.

Weighted network matrix

$$\mathrm{wto}(v,u) = \sum_{e \in \mathrm{ostar}(v) \cap \mathrm{istar}(u)} w(e)$$

A two mode network $\mathcal{N} = ((\mathcal{U},\mathcal{V}),\mathcal{L},w)$ can be represented with the corresponding *matrix* $\mathbf{UV} = [uv[u,v]]_{u\in\mathcal{U},v\in\mathcal{V}}$ defined as

$$uv[u,v] = \begin{cases} \mathrm{wto}(u,v) & (u,v)\in\mathcal{L} \\ \square & \text{otherwise} \end{cases}$$

We extend the set of weight values with a new zero element $\square$ (no link) with rules $\square + a = a$ and $\square \cdot a = \square$.

In the case $\mathcal{U} = \mathcal{V}$ we obtain a square matrix of an ordinary network.

A network $\mathcal{N}$ is *binary* iff the weight w is constant $w = 1$.

Transpose and total

The *transpose* $\mathbf{UV}^T$ of matrix $\mathbf{UV}$ is a matrix on $\mathcal{U} \times \mathcal{V}$ with entries $uv^T[v,u] = uv[u,v]$. $\mathbf{VU} = \mathbf{UV}^T$
Reversed network $\widetilde{\mathcal{N}} = ((\mathcal{V},\mathcal{U}), \widetilde{\mathcal{L}}, vu)$, $\widetilde{\mathcal{L}} = \{(v,u) : (u,v) \in \mathcal{L}\}$, $vu(v,u) = uv(u,v)$ for $(u,v) \in \mathcal{L}$.

$$\mathrm{wod}_{VU}(v) = \mathrm{wid}_{UV}(v), \ldots$$

In the following, we will often identify networks by their matrices.

The *total weight* of links in the network $\mathcal{N} = (\mathcal{V}, \mathcal{L}, w)$ with matrix $\mathbf{N} = [n[u,v]]$

$$T(\mathbf{N}) = \sum_{(u,v)\in\mathcal{L}} w(u,v) = \sum_{u,v} n[u,v] = \sum_u wod_N(u) = \sum_v wid_N(v)$$

Some network transformations

$S(\mathbf{N})$ transforms a network $\mathbf{N}$ into a simple undirected network with link weights equal to the sum of corresponding arc weights.

$D_0(\mathbf{N})$ removes loops from a network $\mathbf{N}$.

Binarization $\mathrm{bin}(\mathbf{N})$ sets the weight of each link to 1.

Multiplication of networks

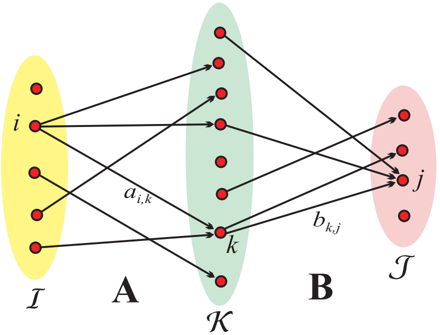
Given a pair of *compatible* networks $\mathcal{N}_A = ((\mathcal{I}, \mathcal{K}), \mathcal{A}_A, w_A)$ and $\mathcal{N}_B = ((\mathcal{K}, \mathcal{J}), \mathcal{A}_B, w_B)$ with corresponding matrices $\mathbf{A}_{\mathcal{I}\times\mathcal{K}}$ and $\mathbf{B}_{\mathcal{K}\times\mathcal{J}}$ we call a *product of networks* $\mathcal{N}_A$ and $\mathcal{N}_B$ a network $\mathcal{N}_C = ((\mathcal{I}, \mathcal{J}), \mathcal{A}_C, w_C)$, where $\mathcal{A}_C = \{(i,j) : i \in \mathcal{I}, j \in \mathcal{J}, c_{i,j} \neq 0\}$ and $w_C(i,j) = c_{i,j}$ for $(i,j) \in \mathcal{A}_C$. The product matrix $\mathbf{C} = [c_{i,j}]_{\mathcal{I}\times\mathcal{J}} = \mathbf{A} * \mathbf{B}$ is defined in the standard way

$$c_{i,j} = \sum_{k \in \mathcal{K}} a_{i,k} \cdot b_{k,j}$$

In the case when $\mathcal{I} = \mathcal{K} = \mathcal{J}$ we are dealing with ordinary one-mode networks (with square matrices).

This definition can be extended to any semiring $(S, +, \cdot, 0, 1)$ [1].

Multiplication of networks



$$c_{i,j} = \sum_{k \in oN_A(i) \cap iN_B(j)} a_{i,k} \cdot b_{k,j}$$

If both networks $\mathcal{N}_A$ and $\mathcal{N}_B$ are binary the value of $c_{i,j}$ counts the number of ways we can go from $i \in \mathcal{I}$ to $j \in \mathcal{J}$ passing through $\mathcal{K}$, $c_{i,j} = |oN_A(i) \cap iN_B(j)|$. oN are out-neighbors, etc.

Derived networks

We can use multiplication to obtain new networks from existing *compatible* two-mode networks. For example, from basic bibliographic networks **WA** (Works, Authors) and **WK** we get

$$\mathbf{AK} = \mathbf{WA}^T \cdot \mathbf{WK}$$

a network relating authors to keywords used in their works, and

$$\mathbf{ACiA} = \mathbf{WA}^T \cdot \mathbf{Ci} \cdot \mathbf{WA}$$

is a network of citations between authors [5].

Networks obtained from existing networks using some operations are called *derived* networks. They are very important in the analysis of collections of *linked* networks.

The weight $AK[a, k]$ is equal to the number of times the author a used the keyword k in his/her works.

The weight $ACiA[a, b]$ counts the number of times a work authored by the author a cites a work authored by the author b; or, more simply, how many times the author a cited the author b.

. . . and projections

Two-mode matrices $\mathbf{UV}$ and $\mathbf{UV}^T$ are always compatible on both sides. The corresponding derived one-mode matrices are called *projections*

- *row* projection $\mathbf{VV} = \text{row}(\mathbf{UV}) = \mathbf{UV}^T \cdot \mathbf{UV}$
- *column* projection $\mathbf{UU} = \text{col}(\mathbf{UV}) = \mathbf{UV} \cdot \mathbf{UV}^T$

They are related $\text{row}(\mathbf{UV}) = \text{col}(\mathbf{UV}^T) = \text{col}(\mathbf{VU})$.
Therefore, it is enough to study the properties of one of them.

Fractional approach

For vectors $x = [x_1, x_2, \ldots, x_n]$ and $y = [y_1, y_2, \ldots, y_m]$ their *outer product* $x \circ y$ is defined as a matrix

$$x \circ y = [x_i \cdot y_j]_{n \times m}$$

then we can express the product $\mathbf{C}$ of two compatible matrices $\mathbf{A}$ and $\mathbf{B}$ as the *outer product decomposition*

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} = \sum_k \mathbf{H}_k \quad \text{where} \quad \mathbf{H}_k = \mathbf{A}[\cdot, k] \circ \mathbf{B}[k, \cdot],$$

$\mathbf{A}[., k]$ is the k-th column of matrix $\mathbf{A}$, and $\mathbf{B}[k, .]$ is the k-th row of matrix $\mathbf{B}$.

$$T(x \circ y) = S_x \cdot S_y \qquad \text{where} \qquad S_x = \sum_i x_i$$

Fractional approach

Variations on projection of 2-mode network
V. Batagelj
Networks
Multiplication
Derived networks
Fractional approach
Normalizations
Example
Conclusions
References

In general, each term $\mathbf{H}_k$ in the outer product decomposition has a different value $T(\mathbf{H}_k) = \mathrm{wid}_A(k) \cdot \mathrm{wod}_B(k)$ leading to over-representation of nodes with large values.

$$T(\mathbf{C}) = T(\sum_k \mathbf{H}_k) = \sum_k T(\mathbf{H}_k) = \sum_k \mathrm{wid}_A(k) \cdot \mathrm{wod}_B(k)$$

To make the contributions of all nodes equal, we can apply the *fractional* approach by normalizing the weights: setting $x' = x/S_x$, we get $S_{x'} = 1$ and therefore $T(x' \circ y') = 1$.

Let $\mathbf{Co} = \mathrm{row}(\mathbf{UV})$ – the *co-occurrence* network.

$$Co[v,z] = Co[z,v] \qquad T(\mathbf{H}_u) = \mathrm{wod}(u)^2$$

$$T(\mathbf{Co}) = \sum_{u \in \mathcal{U}} T(\mathbf{H}_u) = \sum_{u \in \mathcal{U}} \mathrm{wod}(u)^2$$

Normalizations

Stochastic, Markov: $\hat{\mathcal{U}} = \{u \in \mathcal{U} : \text{wod}_{UV}(u) > 0\}$

$$n(UV)[u,v] = \begin{cases} \dfrac{uv[u,v]}{\text{wod}_{UV}(u)} & u \in \hat{\mathcal{U}} \\ \square & u \notin \hat{\mathcal{U}} \end{cases} \qquad \text{wod}_{n(UV)}(u) = \begin{cases} 1 & u \in \hat{\mathcal{U}} \\ 0 & u \notin \hat{\mathcal{U}} \end{cases}$$

$$T(n(\mathbf{UV})) = \sum_{u \in U} \text{wod}_{n(UV)}(u)^2 = \sum_{u \in \hat{\mathcal{U}}} 1 = |\hat{\mathcal{U}}|$$

Strict, Newman: loops excluded

$\text{wod}'_{UV}(u) = \sum_{v \neq u} uv[u,v] = \text{wod}_{UV}(u) - uv[u,u]$

$U' = \{u \in \mathcal{U} : \text{wod}'_{UV}(u) > 0\}$

$$s(UV)[u,v] = \begin{cases} \dfrac{uv[u,v]}{\text{wod}'_{UV}(u)} & u \in \mathcal{U}' \\ \square & u \notin \mathcal{U}' \end{cases} \qquad \text{wod}_{n(UV)}(u) = \begin{cases} 1 & u \in \mathcal{U}' \\ 0 & u \notin \mathcal{U}' \end{cases}$$

$$T(s(\mathbf{UV})) = \sum_{u \in U} \text{wod}_{s(UV)}(u)^2 = \sum_{u \in \mathcal{U}'} 1 = |\mathcal{U}'|$$

Normalizations

Weight-preserving normalization

$$p(UV)[u,v] = \begin{cases} \dfrac{uv[u,v]}{\sqrt{\mathrm{wod}_{UV}(u)}} & u \in \hat{\mathcal{U}} \\ \square & u \notin \hat{\mathcal{U}} \end{cases}$$

$$wod_{p(UV)}(u) = \begin{cases} \sqrt{\mathrm{wod}_{UV}(u)} & u \in \hat{\mathcal{U}} \\ 0 & u \notin \hat{\mathcal{U}} \end{cases}$$

$$T(u) = wod_{p(UV)}(u)^2 = \mathrm{wod}_{UV}(u)$$

[2]

Weighted projections

Fractional projection

$$\mathbf{Cn}_N = S(n(\mathbf{N})^T \cdot n(\mathbf{N}))$$

$\mathbf{Cn}_N^T = \mathbf{Cn}_N$ and $T(\mathbf{Cn}_N) = |\hat{\mathcal{U}}|$.

Strict fractional projection

$$\mathbf{Cs}_N = S(D_0(s(\mathbf{N})^T \cdot n(\mathbf{N})))$$

$\mathbf{Cs}_N^T = \mathbf{Cs}_N$ and $T(\mathbf{Cs}_N) = |\mathcal{U}'|$.

Weight-preserving projection

$$\mathbf{Cp}_N = S(p(\mathbf{N})^T \cdot p(\mathbf{N}))$$

$\mathbf{Cp}_N^T = \mathbf{Cp}_N$ and $T(\mathbf{Cp}_N) = T(\mathbf{N})$. The total of weights in the original two-mode network is distributed over arcs of the projection.

Projections through an intermediate relation

Assume that a network $\mathbf{R}$ on $\mathcal{U} \times \mathcal{U}$ is given. The network

$$\mathbf{Q} = n(\mathbf{UV})^T \cdot \mathbf{R} \cdot n(\mathbf{UV})$$

links nodes from $\mathcal{V}$ *through* the network $\mathbf{R}$.

It holds

- If $\mathbf{R}$ is symmetric, $\mathbf{R}^T = \mathbf{R}$, then also $\mathbf{Q}$ is symmetric, $\mathbf{Q}^T = \mathbf{Q}$.
- $T(\mathbf{Q}) = \sum_{u \in \hat{\mathcal{U}}} \sum_{t \in \hat{\mathcal{U}}} r[u,t] = \hat{T}(\mathbf{R})$.

Therefore, if $\hat{\mathcal{U}} = \mathcal{U}$ then $T(\mathbf{Q}) = T(\mathbf{R})$. The total weight of $\mathbf{R}$ is redistributed over the links of $\mathbf{Q}$.

Projections through intermediate relation

node property

Let $c : \mathcal{U} \to \mathbb{R}$ be a property of nodes. Setting $\mathbf{R} = \text{diag}(c)$ we get

$$T(\mathbf{Q}) = \sum_{u \in \hat{\mathcal{U}}} c(u)$$

The contribution of each active node $u \in \hat{\mathcal{U}}$ to the weights of $\mathbf{Q}$ is $c(u)$.

For example, for the bibliographic network **WA** the property c can be

- the impact factor of the journal in which the work u was published
- the number of references to the work u
- the size (number of characters or words) of the work u

Fractional citations between authors

From the OpenAlex bibliographic database, we obtained in July 2026 a collection of networks (**Ci**, **WA**, **WJ**, **WK**, **WC**) based on works with the keyword `social-network` or `social-network-analysis` and the works cited from them – $|\mathcal{W}| = 956445$ and $|\mathcal{A}| = 159307$. We computed the network of *fractional citations between authors*

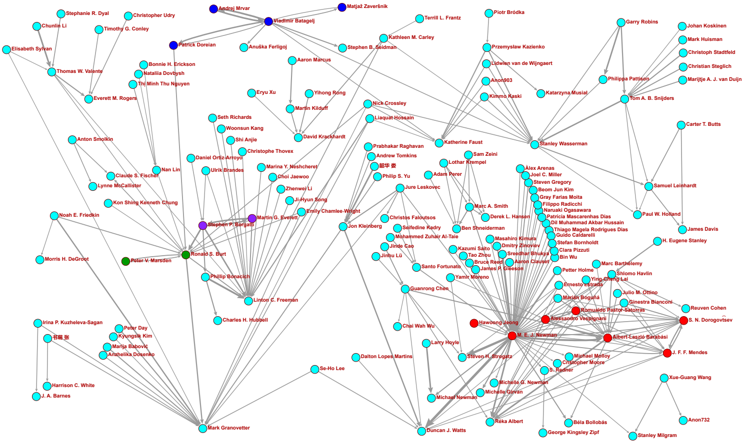
$$\mathbf{ACiAn} = D_0(n(\mathbf{WA})^T \cdot n(\mathbf{Ci}) \cdot n(\mathbf{WA}))$$

The network **ACiAn** is a projection through the normalized citation network $n(\mathbf{Ci})$. Therefore $T(\mathbf{ACiAn}) \leq T(n(\mathbf{Ci})) = |\hat{\mathcal{W}}|$. Each active work contributes value 1 that is distributed over the weights of the network **ACiAn**.

We made a link cut at level 0.4 (component size distribution: 156: 1, 8: 2, 7: 1, 5: 1, 4: 3, 3: 36, 2: 57). We present the main component.

Fractional citations between authors

link cut at level 0.4, main component



almost acyclic – small strong components

Conclusions

- The obtained projections can be analyzed using standard network analysis methods [3, 7, 6].
- This is a traditional recipe for analyzing two-mode networks. Often, the weights are not considered in the analysis, and when they are, we have to be very careful about their meaning. In cases where the direct fractional approach is not meaningful, we can apply the approach proposed for normalized bibliographic coupling in [4, pp. 631–632].
- Traditional projections were defined for binary networks. We extended the definition to weighted networks and proposed some additional options.

Acknowledgments

The computational work reported in this presentation was performed using R and the program for analysis and visualization of large networks, Pajek [8]. The code and data are available at GitHub/bavla [9].

This work is supported in part by the Slovenian Research Agency (research program P1-0294 and research project J5-4596), and prepared within the framework of the COST action CA21163 (HiTEc).

References I

Variations on projection of 2-mode network

V. Batagelj

Networks

Multiplication

Derived networks

Fractional approach

Normalizations

Example

Conclusions

References

- Batagelj, Vladimir: Semirings for social networks analysis, The Journal of Mathematical Sociology, 19(1994)1, 53-68.
- Batagelj, Vladimir, Mrvar, Andrej: Density based approaches to network analysis; Analysis of Reuters terror news network. Workshop on link analysis for detecting complex behavior (LinkKDD2003). cmu/dunja.
- Batagelj, V., Doreian, P., Ferligoj, A., Kejžar, N.: Understanding Large Temporal Networks and Spatial Networks: Exploration, Pattern Searching, Visualization and Network Evolution. Wiley, October 2014.
- Batagelj, V. On fractional approach to analysis of linked networks. Scientometrics 123, 621–633 (2020). https://doi.org/10.1007/s11192-020-03383-y
- Batagelj, V, Cerinšek, M: On bibliographic networks. Scientometrics 96 (2013) 3, 845-864.
- Batagelj, Vladimir. Analysis of the Southern women network using fractional approach. Social Networks 68 (2022): 229-236.

References II

- Batagelj, V, Maltseva, D: Temporal bibliographic networks. Journal of Informetrics, Volume 14, Issue 1, February 2020, 101006.
- Maltseva, D, Batagelj, V: Social network analysis as a field of invasions: bibliographic approach to study SNA development. Scientometrics, 121(2019)2, 1085-1128
- Pajek. http://mrvar.fdv.uni-lj.si/pajek/
- https://github.com/bavla/NormNet/tree/main/TwoMode